



Figure 2: Performance comparison of model links across four targets (Age, Gender, Face, Gender) for D-Student and E-Student. The chart shows training data ratio (%) and target age for both models.

Table 2: IoU scores of model links targeted to the Person model and the Pearson correlations.

performance The model link from Face to Gender achieves 921% accuracy And for model links Gender-to-Age model achieves 30 MAE Compared with student models trained via knowledge distillation on the target models, model links achieves higher prediction performance, especially when the amount of training data is small But for speech I don't understand your meaning, please re-enter with the highest confidence for the localization model, we checked the bounding box whether it is empty, and set it as a label, set its value to 0 or 1. We use the regression coefficient as a label, and skip the two sequence generation models. Table 2 shows the results of the target model link. We can see that the model link performance is positively correlated with the Pearson correlation coefficient. For a target model, we constructed multiple model links from different source models. Then we used the same optimizer and the same hyperparameters to train all source model sets. Tab 3 shows the results of using model link and ensemble model on five target models.

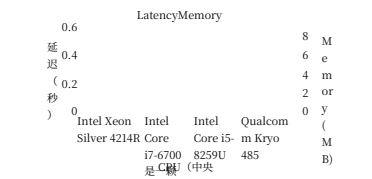


Figure 3: Latency and Memory usage for different devices.

所有的训练样本（48%的比例）模型链接组合优于每一个单独的源模型。我们可以看到有两种典型的案例：主导和互助对于Action, Face, Person目标, Caption, Person, Face来源分别主导了整个集合的表现。但是对于Age和Gender目标, 源模型相互辅助并通过集合实现性能提升。视频分析模型链接

我们对智能建筑系统中的58个摄像头和城市交通监测平台上10个摄像头的48小时视频进行了MLink测试。我们利用数据的前10%进行模型链接和ensemble的训练。我们设置时间长度为一个小时使用初始1%数据进行特征激活概率建模。对于计数模型, 输出精度通过检查预测数的绝对误差是否小于0.5来计算。每个机器学习模型的时间成本是通过训练数据的离线轮廓平均推理时间计算的。智能建筑系统中, 动作/人物/物体模型每帧耗时30/44/60毫秒。在交通监控系统中, 交通/人物/车辆模型每帧耗时55/66/70毫秒。每个ML模型的GPU内存成本是高峰年龄: 46 GB 用于个人计数, 15 GB 用于动作/交通分类, 37 GB 用于对象/车辆计数。我们设定预算B为最大GPU内存分配给ML模型, 以评估MLink如何提高多模型推理的资源效率。我们把每个ML模型的输出作为输入准确率相等并报告他们的平均输出准确率在GPU内存预算下, 基本的“Standalone”仅选择平均耗时最短的模型。我们重复了调度实验三次, 并在表4中报告了结果。由于标准偏差较小(< 0.1), 为了简单起见, 我们没有进行呈现。在两个系统中MLink在输出精度方面优于替代方案。与“Standalone”相比, 在智能建筑系统中, MLink节省了667%的推理执行时间MLink在保持941%输出精度的同时, 实现了对四种不同设备（一台云服务器、一台边缘服务器、一台笔记本电脑和一台移动电话）的部署, 并对其延迟和内存占用进行了测试。如图3所示, MLink只会引入微小的额外开销